

Uniform convergence for dependent Gaussian Fourier series and a sharp cotype obstruction

Candidate full answers to two questions in arXiv:2103.09579

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Status: candidate full results, likely valid. Under exactly the hypotheses of Theorem 4 in the source, condition (8) does imply almost-sure uniform convergence. In contrast, the suggested cotype-2 extension to general stationary Gaussian processes is false, even within the same covariance-matrix class.

1. Source questions

Let $(\xi_n)_{n \in \mathbb{Z}}$ be a centered stationary proper complex Gaussian sequence of unit variance, with

$$\mathbb{E}(\xi_n \overline{\xi_m}) = \gamma(n - m), \quad \mathbb{E}(\xi_n \xi_m) = 0.$$

Theorem 4 of Mukeru's paper assumes that, for some $b \geq 0$, the matrix

$$A_{nm} = |\gamma(n - m)| |nm|^{-b}, \quad n, m \neq 0, \tag{1}$$

defines a bounded operator on $\ell^2(\mathbb{Z} \setminus \{0\})$. For $f \in L^2(\mathbb{T})$ it imposes

$$\sum_{|n| \geq 2} \frac{1}{|n| \sqrt{\log |n|}} \left(\sum_{|k| \geq |n|} |\widehat{f}(k)|^2 |k|^{2b} \right)^{1/2} < \infty. \tag{2}$$

The source proves almost-sure boundedness of the associated random Fourier series. Remark 2 asks whether (2) gives almost-sure uniform convergence, as in the independent case. It also asks for an analogue of Pisier's cotype-2 theorem for general stationary Gaussian processes [1].

2. Proof intuition

The source compares the limiting dependent process with an independent Gaussian Fourier series and concludes boundedness. To obtain convergence, one must compare more at once: put every difference of every two tail partial sums into a single Gaussian process. The matrix hypothesis controls arbitrary finitely supported coefficient vectors, so it controls every canonical-metric increment of this much larger process. Gaussian comparison then bounds the full dependent tail diameter by the independent one.

The independent comparator converges uniformly by the classical criterion already used in the source. Its tail diameter tends to zero in expectation. The dependent tail diameters decrease with the starting index, so expectation tending to zero makes their almost-sure limit zero. This supplies precisely the missing uniform-Cauchy conclusion.

that $a + b + c > 1$ and $x = (x_n)_{n \in \mathbb{Z}}$ with $x_n = |n|^{-c}$ for $n \neq 0$ and $x_0 = 1$. It is the case that

$$\forall n \sum_{m \in \mathbb{Z}} a_{nm} x_m \leq K x_n \quad \text{and} \quad \forall m \sum_{n \in \mathbb{Z}} x_n a_{nm} \leq K x_m.$$

Hence Schur's test yields that the operator defined by (a_{nm}) is bounded. In particular the covariance function γ_H of the fractional Gaussian noise of index H is such that the corresponding matrix (a_{nm}) given by $a_{nm} = \gamma_H(n-m)|nm|^{-b}$ for any $b > 0$ defines a bounded operator. Indeed it is easy to verify that $0 < \gamma_H(n-m) \leq |n-m|^{-2(1-H)}$ for $m \neq n$.

Corollary 1 *Let $f \in L^2(\mathbb{T})$ and $\frac{1}{2} \leq H < 1$. If there exist $\alpha > H$ and $K > 0$ such that*

$$|\hat{f}(n)| \leq K|n|^{-\alpha}, \quad \text{for all } n \neq 0,$$

then the function $t \mapsto \sum_{n \in \mathbb{Z}} \hat{f}(n) \Delta_n e^{2\pi i n t}$ is bounded almost surely. In particular if in addition $f \in C(\mathbb{T})$ but the series $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|$ diverges, then $f \in \mathcal{P}(H)$ but $f \notin A(\mathbb{T})$.

This follows from the fact that by taking $b = H - 1/2 > 0$ it is clear that inequality (8) holds.

Remark 2: It would be interesting to investigate whether inequality (8) implies the uniform convergence (almost surely) of the random Fourier series $t \mapsto \sum_{n \in \mathbb{Z}} \hat{f}(n) \xi_n e^{2\pi i n t}$ as it is the case for i.i.d random variables. Pisier [9] obtained that the Banach space of $f \in L^2(\mathbb{T})$ such that $\sum_{n \in \mathbb{Z}} Z_n \hat{f}(n) e^{2\pi i n t}$ (with (Z_n) i.i.d Gaussian variables) is bounded a.s. is of cotype 2. It would be interesting to consider this question for the general stationary Gaussian process (ξ_n) . Pedersen [7] proved that spectral synthesis holds in $\mathcal{P}$. This question is also of some interest in the general setting discussed in this paper.

Figure 1: The end of Theorem 4, its corollary, and the two questions in Remark 2 on page 8 of arXiv:2103.09579.

The cotype question has a different answer. At the extreme of perfect dependence, ξ_n is the same Gaussian for every n . The corresponding Pisier-type algebra is simply $C(\mathbb{T})$, which contains ℓ_∞^N uniformly and has no finite cotype.

3. Uniform convergence

For symmetric partial sums write

$$S_N(t) = \sum_{|n| \leq N} \hat{f}(n) \xi_n e^{2\pi i n t}.$$

Lemma 1 (coefficient domination). *Let $K = \|A\|$. For every finitely supported complex sequence (c_n) on $\mathbb{Z} \setminus \{0\}$,*

$$\mathbb{E} \left| \sum_n c_n \xi_n \right|^2 \leq K \sum_n |c_n|^2 |n|^{2b}. \quad (3)$$

Proof. Set $u_n = |c_n| |n|^b$. The covariance form is nonnegative, and entrywise absolute values followed by (1) give

$$\mathbb{E} \left| \sum_n c_n \xi_n \right|^2 \leq \sum_{n,m} |\gamma(n-m)| |c_n| |c_m| = \langle Au, u \rangle \leq K \|u\|_2^2. \quad \square$$

Let (g_n) be independent standard proper complex Gaussians and define

$$Y_N(t) = \sum_{|n| \leq N} \widehat{f}(n) |n|^b g_n e^{2\pi i n t},$$

with the harmless zero-frequency convention omitted. Condition (2), split into positive and negative frequencies, is the Marcus–Pisier sufficient condition invoked in the source. Hence (Y_N) converges uniformly on $\mathbb{T}$ almost surely.

Theorem 1 (affirmative answer to Remark 2). *Under the hypotheses of Theorem 4 of the source and condition (2), the partial sums S_N converge uniformly on $\mathbb{T}$ almost surely. In particular the random Fourier series has a continuous version.*

Proof. Define the two tail diameters

$$D_N^S = \sup_{M, L \geq N} \|S_M - S_L\|_\infty, \quad D_N^Y = \sup_{M, L \geq N} \|Y_M - Y_L\|_\infty.$$

We claim

$$\mathbb{E} D_N^S \leq \sqrt{K} \mathbb{E} D_N^Y. \quad (4)$$

First restrict M, L to a finite range and t and $\theta \in [0, 2\pi]$ to finite sets. Consider the centered real Gaussian process

$$X_{M, L, t, \theta} = \operatorname{Re} \left(e^{-i\theta} (S_M(t) - S_L(t)) \right).$$

The difference of any two of its coordinates is a finite linear combination of the ξ_n . Lemma 1 applies to that coefficient vector. Properness gives $\mathbb{E}(\operatorname{Re} Z)^2 = \frac{1}{2} \mathbb{E}|Z|^2$, so its canonical metric is dominated by that of

$$\sqrt{K} \operatorname{Re} \left(e^{-i\theta} (Y_M(t) - Y_L(t)) \right).$$

Sudakov–Fernique comparison proves the finite-index version of (4). Letting the finite range grow and using dense sets of t and θ gives (4); the phase parameter uses $|z| = \sup_\theta \operatorname{Re}(e^{-i\theta} z)$.

Because (Y_N) converges uniformly almost surely, $D_N^Y \rightarrow 0$ almost surely. The symmetric shells of (Y_N) are independent symmetric random elements of the separable Banach space $C(\mathbb{T})$. Its limit Y is Gaussian, so Fernique’s theorem gives $\mathbb{E}\|Y\|_\infty < \infty$. The Banach-valued Lévy maximal inequality then yields

$$\mathbb{E} \sup_N \|Y_N\|_\infty < \infty.$$

Since $D_N^Y \leq 2 \sup_M \|Y_M\|_\infty$, dominated convergence gives $\mathbb{E} D_N^Y \rightarrow 0$. Equation (4) implies $\mathbb{E} D_N^S \rightarrow 0$.

Finally, D_N^S decreases as N increases. Therefore

$$\mathbb{E} \left(\lim_{N \rightarrow \infty} D_N^S \right) = \lim_{N \rightarrow \infty} \mathbb{E} D_N^S = 0.$$

Thus $D_N^S \rightarrow 0$ almost surely, exactly saying that (S_N) is uniformly Cauchy. Its uniform limit is continuous. At each fixed t , the source’s $L^2(\mathbb{P})$ convergence identifies this limit with the stated formal Gaussian Fourier series. $\square$

4. The cotype extension fails

Theorem 2 (counterexample to general cotype 2). *There is a stationary proper complex Gaussian sequence, satisfying the matrix hypothesis (1) for some b , whose Pisier-type algebra has no finite cotype. Hence a general stationary-Gaussian analogue of Pisier’s cotype-2 conclusion is false.*

Proof. Let Z be one standard proper complex Gaussian and set

$$\xi_n = Z \quad (n \in \mathbb{Z}).$$

This sequence is stationary, $\gamma(k) = 1$, and its spectral measure is the point mass δ_0 . In the source’s construction the canonical process is

$$X_f(t) = f(t)Z.$$

Consequently every $f \in C(\mathbb{T})$ belongs to $\mathcal{P}(\delta_0)$, and

$$\|f\|_\infty \leq \|f\|_{\mathcal{P}(\delta_0)} \leq (1 + 2\mathbb{E}|Z|)\|f\|_\infty. \quad (5)$$

Thus $\mathcal{P}(\delta_0)$ is isomorphic to $C(\mathbb{T})$.

For each N , choose N continuous norm-one bumps with pairwise disjoint supports. Their span is isometric to ℓ_∞^N in the supremum norm, and (5) makes the distortion uniform in N . If the space had cotype $q < \infty$ with constant C , its coordinate vectors would give

$$N^{1/q} \leq C \left(\mathbb{E}_\varepsilon \left\| \sum_{j=1}^N \varepsilon_j e_j \right\|_\infty^q \right)^{1/q} = C,$$

an impossibility as $N \rightarrow \infty$.

The example even satisfies (1). If $b > 1/2$, then on nonzero indices

$$A_{nm} = |n|^{-b}|m|^{-b} = v_n v_m, \quad v = (|n|^{-b})_{n \neq 0} \in \ell^2.$$

Thus $A = v \otimes v$ is a bounded rank-one operator. □

5. Scope and novelty

Theorem 1 answers the first question in Remark 2 under exactly the source hypotheses. Theorem 2 answers the adjacent general cotype question negatively, including inside the source’s weighted covariance class. We do not settle the third prompt, spectral synthesis for general $\mathcal{P}(\mu)$; the point-mass example is $C(\mathbb{T})$ and gives no obstruction there.

A bounded search on 13 August 2026 used the exact title and the terms stationary Gaussian Fourier series, uniform convergence, cotype, and Pisier-type algebra. It found the source and classical background, but no later answer to either prompt and no matching all-tail comparison. Novelty confidence is moderate because the affirmative argument is a short maximal upgrade of the source’s own comparison.

Human-review focus. Check that Lemma 1 applies to differences of arbitrary partial sum coordinates; the realification in (4); the integrability of the independent maximal norm; and whether the cotype prompt is intended for precisely $\mathcal{P}(\mu)$ or its equivalent polynomial completion.

References

- [1] S. Mukeru, *A generalisation of Pisier homogeneous Banach algebra*, arXiv:2103.09579 (2021).
- [2] M. B. Marcus and G. Pisier, *Random Fourier Series with Applications to Harmonic Analysis*, Princeton University Press, 1981.
- [3] M. Ledoux and M. Talagrand, *Probability in Banach Spaces*, Springer, 1991.